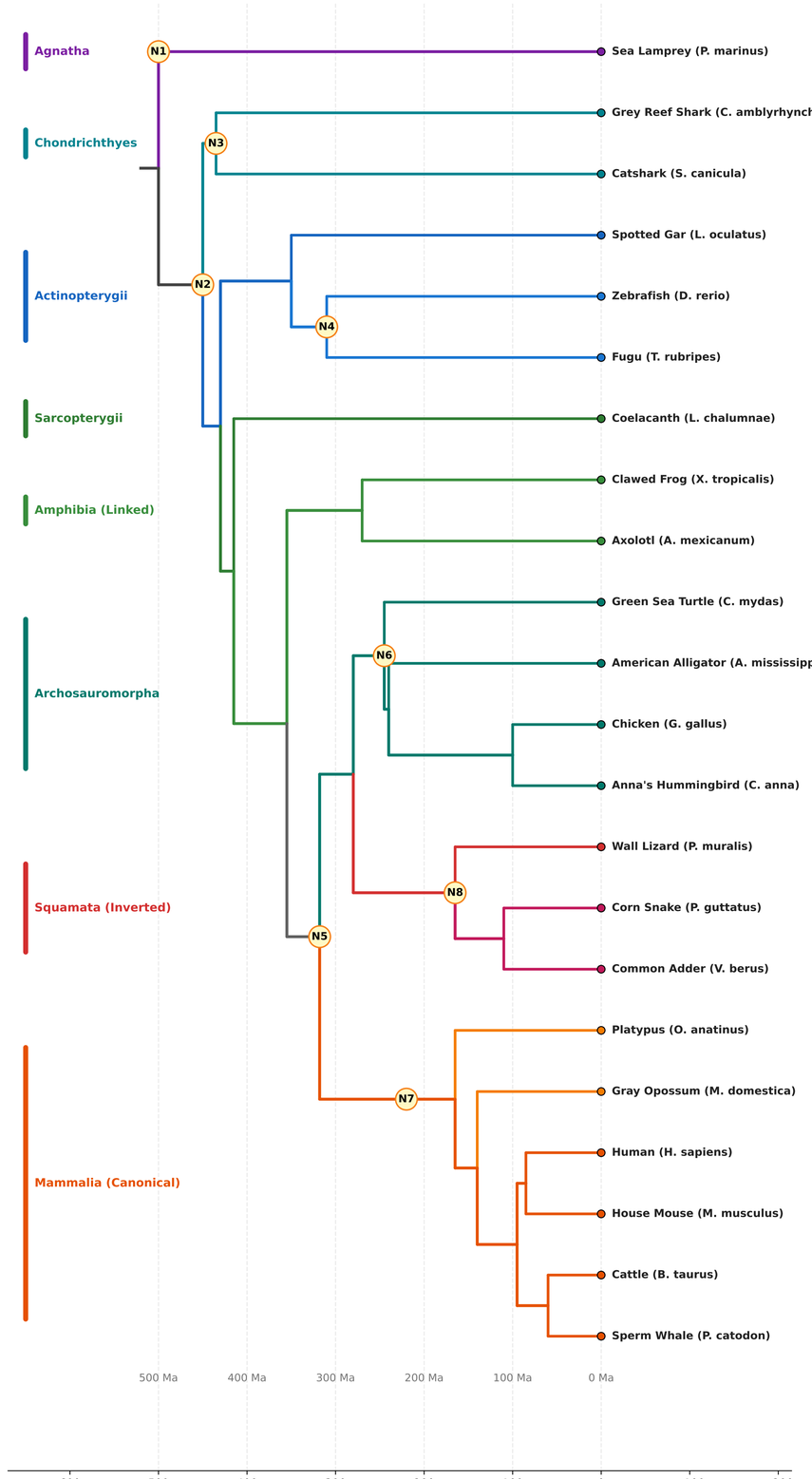
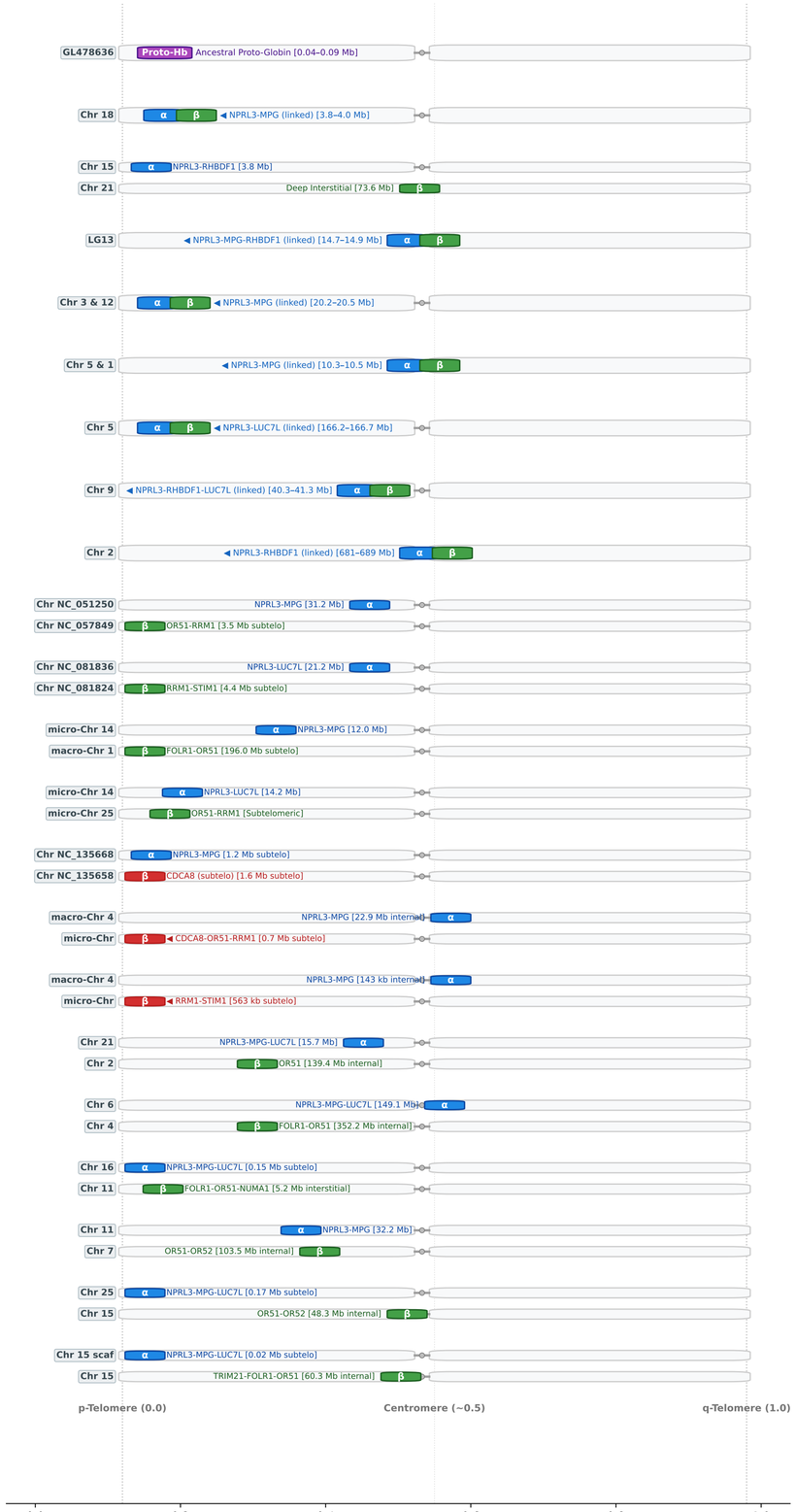
Figure 4: Macroevolutionary Synteny Architecture and Codon Divergence Across Vertebrates

A. Calibrated Vertebrate Tree & Evolutionary Milestones



B. Conserved Syntenic Blocks & Chromosomal Architecture



C. Codon Decomposition (GC1, GC2, GC3) & Subunit Divergence (ΔGC3)



α-Globin GC1
α-Globin GC2
α-Globin GC3
β-Globin GC1
β-Globin GC2
β-Globin GC3 (Canonical/Linked)
β-Globin GC3 (Squamate Inverted)
ΔGC3 > 0 Divergence Bar (α-dominant)
ΔGC3 ≈ 0 Divergence Bar (Balanced)
ΔGC3 < 0 Divergence Bar (β-dominant)